

Statement of Work

Requisition 143041-26-0066

The NWFSC Fish Ecology (FE) Division requires the skills of an experienced C++ Programmer that has applied ecological knowledge to work on the Comprehensive Passage (COMPASS) model. COMPASS produces predictions of the effects of alternative dam and reservoir operations on juvenile salmon and steelhead survival and passage through the Columbia River hydropower system. It is an essential backbone for salmon life cycle modeling and supports decisions about hydropower operations and salmon and steelhead management.

Technical details

COMPASS is a complex multifaceted ecological model written in C++ for maximum computing performance and is expected to remain in use for the foreseeable future. The COMPASS codebase is based on object-oriented design and is split into many .cpp files, with each file containing functions related to handling a specific task. The total codebase is ~230,000 lines. The model is built under the Qt 6 development environment with C++ 17. Boost is also used in the compilation process. The codebase resides in a Git repository, with documented processes to install and configure the Qt environment to compile and deploy builds. Other tools are generally not under consideration to limit dependencies and need for licenses. Windows is the primary platform for the graphical user interface (GUI) and the anticipated userbase (a small team), but the current codebase and development environment can also compile the GUI for Linux and Mac OSX. Individual builds of the model may be in use for ~1 year before being superseded by a new build. NOAA will provide all data necessary to run the model, including input datafiles, historical model builds, and historical model outputs. The contractor will have access to NOAA internal systems, repositories, and servers necessary to accomplish the tasks. The contractor is expected to work with NOAA personnel to troubleshoot and debug the program and to implement new ecological features. Code and documentation developed under this contract will be the property of the U.S. Government.

Key qualifications

Required: A degree in a computer science field, 10+ years of applied programming experience, and proficiency in C++ specifically. Examples of completed projects using C++ can be used to demonstrate proficiency.

Strongly preferred: Coursework in ecology, fisheries, oceanography, natural resources, or a related field. Experience working with complex ecological models that inform management decisions.

Communication and Documentation

NOAA expects regular (weekly to monthly) reports of changes made and any updates to user documentation. NOAA will work closely with the programmer to ensure regular significant progress and be flexible about the proportion of time spent across tasks each month. In practice, this means scheduled team progress meetings, ad hoc meetings, and consistent email or phone communication. The NOAA project leader will coordinate task priority and assess progress. The programmer will document progress and propose any needed changes. Decisions about best ways to solve particular problems will be decided by the team. Written approval by NOAA is required prior to any scope expansion.

Base Year. The following tasks shall be completed by the last date of the base year (2026-2027). Tasks, listed in priority order below, are considered separate billable deliverables. Generally, a deliverable will be a summary describing how the task was performed with supporting documentation giving evidence that the updates work as expected.

- 1) Complete development of the graphical user interface, which allows people to use the model in a user-friendly framework. The GUI is currently functional, with basic features for reading in datafiles, running scenarios, displaying key output statistics, editing certain model parameters, and saving modified scenarios to output files. Missing functionality is primarily related to the ability for the user to modify more of the model parameters, and to edit environmental data such as flow or temperature. The contractor is responsible for creating the Windows installer package for the GUI, following a documented process.
- 2) Complete development of the adult passage module that allows for upstream migration and dam passage. This will include delay at dam tailraces, passage timing through adult fishways, travel time and survival in reservoirs, fallback over dams, and losses due to harvest. Preliminary development of these features is underway. A functionally complete version would include defensible ecological models for salmon reservoir survival and travel time, tailrace delay, ladder movement, fallback, and harvest. Parameters will be calibrated to data using a combination of radio-telemetry data and PIT tag data in collaboration with project scientists.
- 3) Any future modifications necessary to allow the model to provide products requested by the West Coast Region office needed for investigating potential management actions. For example, requests may include specific cases evaluating operations such as spill levels that are being evaluated in support of litigation. Many other examples are also possible to provide real-time answers to support decisions associated with the Magnuson Stevens and Endangered Species Acts. New needs may arise as water resource management in the region changes, which may necessitate additions or modifications to the COMPASS model.
- 4) Routine maintenance and minor modifications necessary to keep the model working and improve model performance and/or functionality. The current COMPASS model runtime is 5 to 30 seconds for a typical model scenario- year. A Monte Carlo process running tens of thousands of scenario-years (the most intensive type of run we do) can take 24-48 hours. Memory footprint is ~200 MB total virtual usage. All current acceptance tests for the COMPASS model are based on parity – i.e., evaluating whether the program produces the correct answer for various test cases. The current performance statistics of the COMPASS model are considered acceptable, but as additional code is added, we need to ensure that it remains efficient to operate. This is a lower priority than implementing ecological features and addressing requests as outlined above.

Option Year 1. The following tasks shall be completed by the last date of option year 1 (2027-2028). Note that option years are funded annually as funds allow.

- 1) Continue to support products developed in year 1, including the GUI and adult passage module.
- 2) Continue to implement modifications requested by the West Coast Regional office.
- 3) Expand the model to include travel time and survival in tributaries. In particular, allow models of early migration that include periods of stationarity and incremental movement before the initiation of active migration during the juvenile life stage. Early migration behaviors vary across populations through space and time. Implement functionality of ecological components concurrently developed by project scientists into the codebase. Parameters will be calibrated to data using a combination of active tag and PIT tag data in collaboration with project scientists.
- 4) Expand additional features in the COMPASS model if any needs are identified by NOAA scientists. Example feature expansions can include: a) modification of the model to allow for the life history of fall Chinook, which can hold in reservoirs over winter; and b) expanding the model to the Columbia River estuary to allow movement and survival through that section of the river while accounting for changes in water velocity due to tidal influence. The specific features are not yet decided and may change given new agency priorities.
- 5) Continue to improve model efficiency and ease of use if it is deemed necessary and if time permits.

Option Year 2. The following tasks shall be completed by the last date of option year 2 (2028-2029). Note that option years are funded annually as funds allow.

- 1) Tasks identified in Option year 1.
- 2) Expand features based on tasks completed in the previous year.